

Santiago Martínez Novoa

Systems and Computing Engineer

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EDUCATION

Universidad de los Andes <i>M.Sc. Information Engineering</i>	Bogotá, Colombia 2025 – 2026 (To be conferred)
Universidad de los Andes <i>B.Sc. Systems and Computing Engineering (Cum Laude), Minor in French</i>	Bogotá, Colombia 2021 – 2024

EXPERIENCE

LLM-Based Applications - Research Assistant <i>Universidad de los Andes</i>	2025-06 – Present Bogotá, Colombia
- Developed comprehensive master's-level RAG course using LangChain, LangGraph, LangSmith, and RAGAS - Developed comprehensive master's-level agent development course using LangGraph and LangSmith	
Graduate Teaching Assistant <i>Universidad de los Andes</i>	2025-06 – Present Bogotá, Colombia
- Supervised 100+ master's students in RAG and agent development courses - Evaluated student performance on practical LLM implementation projects	
Data Scientist - Graduate Research Assistant <i>Vice Dean's Office of Student Affairs, Universidad de los Andes</i>	2025-01 – Present Bogotá, Colombia
- Optimized ETL pipeline performance by 87% by migrating from Pandas to Polars, reducing processing time from 15 to 2 minutes - Developed Power BI dashboards delivering actionable insights on student performance and institutional metrics - Replaced external counseling platform, saving the university ~\$100K USD annually - Co-implemented system alerting over 1000 critical student cases - Implemented complete medallion architecture in Databricks, fully automating institutional data pipelines	
Generative Artificial Intelligence for Research - Research Assistant <i>Universidad de los Andes</i>	2025-01 – 2025-06 Bogotá, Colombia
- Built locally-deployed RAG systems using Ollama and Gemini API for domain-specific research questions - Designed LLM evaluation architecture using Prometheus, BERT score and OpenAI API for response quality assessment - Developed custom AI agents to optimize university research processes - Automated student assessment using LLMs aligned with rubrics and expected answers	
Machine Learning Intern <i>Precia PPV S.A</i>	2024-08 – 2024-12 Bogotá, Colombia
- Developed MLP model reducing financial calculation time by 70% for client applications - Implemented end-to-end AWS infrastructure with automated ETL pipelines using Glue and Lambda - Presented our work on models and AWS infrastructure at Analytics Forum 2025	
Summer Research Intern <i>College of Computing and Information Science, Cornell University</i>	2024-06 – 2024-08 Ithaca, United States
- Designed Data Discovery architecture to explore and modify datasets in various formats - Developed upload and scoring system through semantic search across multiple databases	

TECHNICAL SKILLS

- Programming Languages:* Python, SQL, JavaScript, TypeScript
- Machine Learning & AI:* Scikit-learn, Keras, PyTorch, TensorFlow
- LLM & Natural Language Processing:* Transformers, BERT, LangChain, LangGraph, LangSmith, Ollama
- Data Processing:* Pandas, Polars, Databricks

- *Cloud & DevOps*: AWS, Azure Functions, Azure Blob Storage
- *Databases*: PostgreSQL, MongoDB, ChromaDB, Pinecone